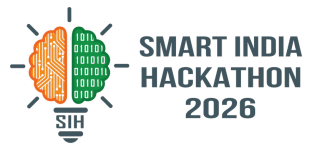


SMART INDIA HACKATHON 2026



Team Name: **UniMinds**

Team Id: **152431**

Problem Statement ID: **SIH 26126**

Problem Statement Title: **Vision Based Autonomous Navigation for Unmanned Ground Vehicle for Outdoor Environment**

Theme: **Smart Automation**

PS Category: **Software**

Team Lead: **Juhil Modi**

Team Members: **Saswata Das, Harshit Tiwari,
Rohan Jangam, Ritesh Prajapati
& Sanskriti**



1 · The Problem & Our Prototype Setup

1 — No GPS, No Map, One Camera

The UGV must reach **Point B** outdoors where GPS is jammed or blocked and no map exists. A LiDAR costs about USD 4,000; one RGB camera costs USD 25 and sees the scene in rich detail.

2 — The Hard Part: Flat Images

A photo has no distance. DRISHTI must turn pixels into distance, terrain height, safe ground, its own motion and a stop decision. Our 12 cm camera height, 4.5 cm clearance and 3 cm step limit are software settings, so a new vehicle needs no retraining.

2 · How DRISHTI Meets the Three PS Challenges

Path Detection

Depth, a PIDNet-S terrain model and a 17-channel head mark each region **safe, risky, obstacle or unknown**.

- Seven classes: trail, grass, rough vegetation, obstacle, water, sky, dynamic.
- Judged against the vehicle's own clearance and step limits.

Visual Localization

Corner features tracked frame to frame, filtered by MAGSAC and scaled by the depth estimate.

- Rolling 2.5D map of 6 cm cells: height, risk, confidence, age.
- Place recognition closes loops; the rover's built-in IMU fixes scale and drift.

Collision Avoidance & A→B

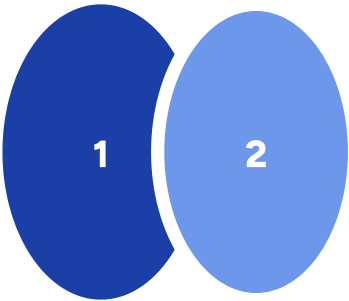
Arcs head straight for **Point B** on open ground. Boxed in? **D* Lite** replans a way around over the ground seen so far; unseen ground costs, never free.

- A person or animal steps in: dynamic cells are inflated and expire in under 1 s, forcing a replan.
- Safety gate on every command: **GO, SLOW, REROUTE or STOP**.

3 · Why This Architecture Holds Up

Vehicle Geometry as Parameters

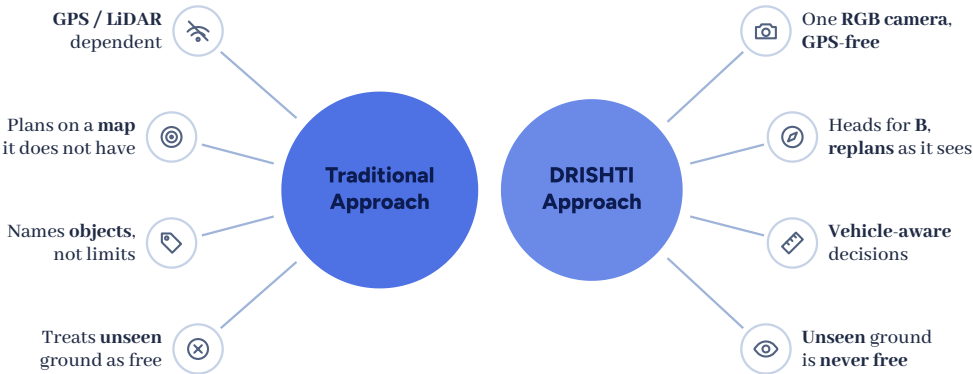
Camera height, clearance and step limit are software settings, so the same visual evidence can be judged for a different vehicle without retraining. Where the ground fit is unreliable, the region is marked **invalid** and treated as unknown.



Learned Suggests, Rules Decide

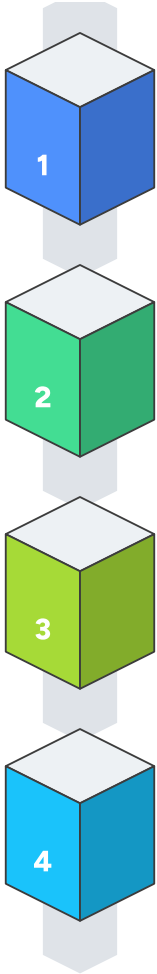
A PPO policy trained inside our GRU world model drove further, but on 60 held-out states it collided 98.3% of the time (always-forward: 60%). It had exploited its own simulator. So learning only suggests; the rule-based supervisor decides, and overrode it on 933 of 1,500 frames.

What Makes Our Solution Unique



Model Architecture

- **Input Layer**
 - One RGB camera - the only world sensor
 - Five ten-second clips, daylight to low light
 - No GPS, LiDAR or prior map; on the rover, a built-in IMU senses self-motion only
- **Processing Engine**
 - Depth Anything V2 + ground-plane fit per frame: shift fixed to 0, SVD + Cauchy IRLS
 - 12 cm camera height is a stated assumption, not a calibration
 - PIDNet-S, 7.7M params - seven terrain classes
 - 17-channel MobileNet head - safe / risky / obstacle / unknown
 - Trust score: 0.48 to 0.27 in low light, not a collision probability
- **Navigation Analysis**
 - Depth-anchored VO - ORB features + MAGSAC
 - 2.5D map, 6 cm cells aging to UNKNOWN; height-step filter: slope vs kerb
 - 17 arcs head for Point B; D* Lite replans around dead ends, then an ordered safety gate



Project Lifecycle

-  **Data Preparation**

Five ten-second outdoor clips, daylight to low light. No human labels: targets come from a teacher model and vehicle geometry.
-  **Model Training**

PIDNet-S distilled from a SegFormer teacher: 66.83% mIoU and 89.64% pixel agreement. World model and PPO trained on logged maps.
-  **Integration**

One Python module, depth to supervisor, wrapped as a ROS 2 node (camera in, cmd_vel out) for Gazebo Harmonic and the rover.
-  **Validation**

45 verified videos across nine of eleven stages. Terrain stage: 46 ms a frame on a laptop RTX 4050, measured.

Technology Stack

Language & Core

Python

NumPy

Libraries

PyTorch

OpenCV

ONNX Runtime

SB3 (PPO)

Models

Depth Anything V2

PIDNet-S

SegFormer-B0

MobileNetV3

Navigation & Deployment

GRU World Model

ROS 2 Jazzy

Gazebo

Jetson Orin

 View our complete codebase github.com/X-DIABLO-X/DRISHTI

The Industry Has Already Chosen Cameras

- 1 — 2021 • Tesla drops radar
Model 3 and Y switch to Tesla Vision - cameras only.
- 2 — 2022 • Ultrasonics removed
Parking and active safety move to cameras. Safety ratings held or improved.
- 3 — 2022 onward • Optimus humanoid
Tesla's robot runs the same camera-driven neural stack as FSD.
- 4 — 2026 • One camera matches LiDAR
Depth Anything V2 + mono-inertial SLAM hit the same success rate as a 128-beam LiDAR in real trials.
- 5 — NOW • DRISHTI
The same bet, on the ground: one camera, off-road, GPS-denied.

Cheaper, Lighter, and Accurate Where It Counts

Production Cost

~15x cheaper

camera + Jetson vs one 3D LiDAR alone; the whole rover is about USD 365

USD 4,000

USD 274

16-beam LiDAR (VLP-16)

Camera + Jetson Orin Nano Super

Battery & Reliability

7 - 25w

the Jetson-class power envelope our models are sized for

7.7M terrain net params

8 MB after INT8

46 ms terrain stage, laptop GPU

Less power on sensing means more battery left for driving, and fewer parts to fail in the field.

Accuracy

"Tesla Vision vehicles have either maintained or improved their active safety ratings" after radar was removed.

TESLA • TRANSITIONING TO TESLA VISION

Depth foundation models now learn from 62M+ images. DRISHTI adds what a point cloud lacks: a vehicle-aware safe / risky / obstacle / unknown call for every 6 cm cell.

Cheap Enough to Lose. Cheap Enough to Swarm.

Defence

Attritable by design: losing a unit is a line item, not a crisis. The idea behind "small, smart, cheap, and many".

Patrol

Swarms on borders and perimeters: many low-cost units cover more ground than one costly vehicle.

Rescue

Send a swarm into a landslide zone before people. Every unit that finds someone has paid for itself.

Agriculture

Priced for farmers, not fleets: navigation under crop canopy without a LiDAR bill.

Consequences & Merits

What DRISHTI Does Better



Jam-proof by construction

No GNSS in the loop, so jamming and spoofing have nothing to attack.



Vehicle-aware, not label-aware

Judges ground against this chassis: **4.5 cm** clearance, **3 cm** step. New robot, **no retraining**.



Never guesses

Unseen ground stays **UNKNOWN**, **never free space**. The vehicle slows or reroutes instead.



Two independent safety paths

A learned risk map plus hard geometric rules. Either one alone can stop the vehicle.



Every decision explained

Every decision stores the **rule that fired** and a **human-readable reason**. No black box.



Edge-ready software

INT8 terrain net: **8 MB**. Terrain stage: **46 ms** a frame on a laptop GPU; sized for **Jetson**.

By the Numbers

0

GNSS **signals** in the control loop: nothing to jam, nothing to spoof

62%

933 of 1,500 frames where the supervisor replaced the policy's action

44%

measured confidence drop in low light; weak evidence slows the vehicle

4×

smaller terrain model after INT8, at **92%** agreement

DRISHTI is built to drive from **Point A to Point B** in Gazebo and on a **USD 365** rover. Its perception is measured on recorded footage: it turns incomplete visual evidence into a **cautious motion decision**.

What Changes When It Deploys



People out of the first metre of danger

Soldiers and rescuers send a unit ahead instead of walking onto unknown ground.



Autonomy a budget can buy

A **USD 25** camera stands in for a **USD 4,000** LiDAR, so fleets and swarms become affordable.



Value that stays in software

Commodity camera, any edge GPU. The hard part is software that can be built and maintained in India.



Trust you can audit

Rule-level logs let operators, testers and certifiers see exactly why the vehicle stopped.

From Footage to Field



Drift-proof localization

The rover's **built-in IMU** fixes metric scale; place-recognition **loop closures** feed a **GTSAM** pose graph.



Point A to Point B, closed loop

Arcs head for Point B; **D* Lite** replans around dead ends every cycle. Driven in **Gazebo Harmonic**, scored on success rate, collisions and interventions.



On a real rover

USD 90 WAVE ROVER + **Jetson Orin Nano Super** + one camera: about **USD 365** all in.

 One RGB frame



STAGE 1

Depth & Scale

Yang, L., et al. (2024) [NeurIPS 2024](#) · [arXiv 2406.09414](#) ↗

[Depth Anything V2](#)

HOW WE USE IT Gives us relative depth. We run it frozen.

Marsal, R., et al. (2025) [IROS 2025](#) · [arXiv 2412.14103](#) ↗

[A Simple yet Effective Test-Time Adaptation for Zero-Shot Monocular Metric Depth](#)

HOW WE USE IT Their scale idea, anchored on the ground instead of an IMU.



STAGE 2

Terrain

Xu, J., et al. (2023) [CVPR 2023](#) · [arXiv 2206.02066](#) ↗

[PIDNet: A Real-time Semantic Segmentation Network Inspired by PID Controllers](#)

HOW WE USE IT Our 7.7M-param PIDNet-S, written from scratch.

Hinton, G., et al. (2015) [arXiv 1503.02531](#) ↗

[Distilling the Knowledge in a Neural Network](#)

HOW WE USE IT Teacher to student: 66.83% mIoU, 89.64% pixel agreement.



STAGE 3

Traversability & Trust

Howard, A., et al. (2019) [ICCV 2019](#) · [arXiv 1905.02244](#) ↗

[Searching for MobileNetV3](#)

HOW WE USE IT Backbone of our 17-channel traversability model.

Gal, Y. & Ghahramani, Z. (2016) [ICML 2016](#) · [arXiv 1506.02142](#) ↗

[Dropout as a Bayesian Approximation](#)

HOW WE USE IT Eight dropout passes tell us how unsure the model is.



STAGE 4

Motion & Memory

Campos, C., et al. (2021) [IEEE T-RO 2021](#) · [arXiv 2007.11898](#) ↗

[ORB-SLAM3: Visual, Visual-Inertial and Multi-Map SLAM](#)

HOW WE USE IT Our front end is ORB-SLAM3-style, without IMU or loop closure.

Marsal, R., et al. (2026) [arXiv 2604.03096](#) ↗

[An Open-Source LiDAR and Monocular Off-Road Autonomous Navigation Stack](#)

HOW WE USE IT Our blueprint: mono depth + VINS scale, 2.5D map, A* to the goal.



STAGE 5

Planning & Safety

Ha, D. & Schmidhuber, J. (2018) [NeurIPS 2018](#) · [arXiv 1803.10122](#) ↗

[World Models](#)

HOW WE USE IT Our GRU world model: 8-channel map to a 96-number state.

Schulman, J., et al. (2017) [arXiv 1707.06347](#) ↗

[Proximal Policy Optimization Algorithms](#)

HOW WE USE IT Our PPO policy: suggestion only, 98.3% collisions reported.